$$J_+ + J_- = -D\,\frac{d(c_+ + c_-)}{dx} + \mu F(c_+ - c_-)E_x$$

$$J_\Sigma \equiv J_+ + J_-$$

$$J_\Sigma = -D\,\frac{dC}{dx} + \mu F(c_+ - c_-)E_x, \qquad C \equiv c_+ + c_-$$